COLOR COURIER

Autonomous color sorting with RGB perception and A* planning

1. Project overview and objective

The robot sorts three parcels (red, blue, and yellow) into matching bins while navigating around obstacles. The task combines visual classification, route planning, motion control, and pick-and-place state transitions in one reproducible simulation.

2. Method and design choices

COMPONENT	IMPLEMENTATION	DESIGN CHOICE
Color perception	Nearest Euclidean distance from simulated camera RGB to three color prototypes.	Gaussian observation noise (SD 7); predicted class and confidence are shown.
Route planning	A* search on an 8-connected occupancy grid.	Inflated obstacle boundaries; replan for each parcel and destination.
Motion and task	Simplified planar differential-drive kinematics with waypoint following.	Pickup and drop are state transitions; all stages are annotated in the video.

3. Results

All three parcels were classified and delivered to their matching bins. Classification confidence was 0.94 for blue, 0.89 for yellow, and 0.93 for red. The demo is 120 seconds at 12 fps; no speed-up was applied.

4. Limitations and next steps

This is a custom 2D visualization simulation, not a ROS, Gazebo, Genesis, or physical-robot run. The classifier assumes three known colors. Grasping is represented by task-state transitions, so contact dynamics, occlusion, friction, wheel slip, and real-world safety are not evaluated. Next steps include image-based object detection, robustness to lighting changes, dynamic-obstacle replanning, and validation in a physics simulator or on hardware.

5. Deliverables and reproduction

Source code and instructions are in the accompanying repository package. Run the script with Python, NumPy, Pillow, and ffmpeg to reproduce the MP4. Before submission, replace the account-ID placeholder in the report filename with the actual OmniCampus account ID.

No personal name, affiliation, or email address is included.